

M Aditya Kiran

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Education

East West Institute of Technology, Bengaluru

Bachelor of Technology in Computer Science

CGPA: 8.9

Technical Skills

AI/ML:	TensorFlow, scikit-learn, Pandas, Retrieval-Augmented Generation (RAG), LLMs (Ollama), Feature Attribution
Languages & Core:	Python (Advanced), Javascript, Node.js, SQL
Backend & Cloud:	FastAPI, AWS (EC2, S3), Docker, Google Cloud Vision API
Tools & Frameworks:	Git, Postman, SQLite, Streamlit, Pydantic

Experience

Machine Learning Engineer (Intern)

Feb 2026 – May 2026

International Institute of Medical Science and Technology Council

- Deploying YOLO26s with a Scale-Targeted Attention Learning (STAL) backbone to detect hairline cracks and efflorescence, explicitly tuned for 16GB VRAM edge hardware.
- Replaced standard Non-Maximum Suppression (NMS) with a bipartite matching head to remove inference bottlenecks, stabilizing processing at 18.1ms for real-time field diagnostics.
- Shifted defect localization from AABB to Oriented Bounding Boxes (OBB) using ProbIoU loss, slashing background noise intrusion from 68% to 14% by modeling boxes as 2D Gaussians.
- Integrated gradient-free Eigen-CAM hooks into the backbone's C2f layer, enabling field engineers to objectively verify model predictions via real-time heatmap activations.
- Refined the training pipeline by injecting 841 hard-negative examples (shadows, construction joints, and formwork marks), effectively neutralizing false positives in high-clutter environments.

AI/ML Engineer (Apprentice)

Oct 2025 – Feb 2026

TNS Foundation (Mitsubishi UFJ Financial Group - MUFG Track)

- Developed clinical risk-stratification models for heart disease diagnosis, achieving 0.92 ROC-AUC by benchmarking ensemble classifiers (Random Forest, SVM) against Logistic Regression baselines.
- Engineered industrial efficiency predictors for machinery maintenance, automating feature extraction pipelines to map operational variance to equipment failure thresholds.
- Standardized hyperparameter optimization and k-fold cross-validation workflows to ensure model reproducibility across heterogeneous clinical and industrial datasets.

Projects

AI-Powered Nutritionist | Full-Stack ML Developer

Tech Stack: Flutter, FastAPI, Docker, Google Cloud Vision API, LLM (Ollama)

- Architected a microservice-based application leveraging Google Cloud Vision API to estimate calorie intake from food images, achieving ~85% accuracy.
- Developed a Retrieval-Augmented Generation (RAG) pipeline using Ollama to deliver personalized dietary recommendations, reducing hallucination rates by 30%.
- Built and deployed REST APIs using FastAPI, containerized with Docker, and automated CI/CD

pipelines.

Algorithmic Stock Trading Dashboard | Data Engineer

Tech Stack: Python, Streamlit, scikit-learn, SQLite, Pydantic

- Built a real-time analytics dashboard using Streamlit to monitor portfolio volatility, sector exposure, and financial risk metrics.
- Implemented anomaly detection models using scikit-learn to identify irregular price movements based on historical variance.
- Engineered robust data validation pipelines using Pydantic, ensuring high-integrity financial data processing with sub-200ms latency.

Heart Disease Risk Prediction System | ML Engineer

Tech Stack: Python, scikit-learn, Streamlit, Docker

- Developed a binary classification model using 13 clinical indicators; benchmarked Logistic Regression, Random Forest, and SVM models.
- Optimized model performance using k-fold cross-validation and GridSearchCV, achieving ROC-AUC of 0.92 with recall prioritization.
- Deployed a containerized Streamlit-based decision support interface for fast and interpretable clinical predictions.

Verified Professional Credentials

AWS Academy Graduate – Cloud Foundations <i>Amazon Web Services (AWS) Training Badge</i>	Aug 2025
Cyber Security Foundation Certification ID: ETHEWI00006646 <i>Ethnotech Academy</i>	Dec 2024
Core Java Programming ID: ETHEWI00003026 <i>Ethnotech Academy</i>	Nov 2024
Data Science Using Python ID: ETHEWI00004828 <i>Ethnotech Academy</i>	Aug 2024
Full-Stack Web Technologies ID: ETHEWI00001209 <i>Ethnotech Academy</i>	Sep 2023